Chess Blunder Predictor

Chess Blunder-Risk Analyzer | Braden Hayes and Nate Poppe | Baseline: October 28, 2026

1. Task List - Assignment #5

Four graded components followed by the work breakdown structure and risk register. Canvas calls this combined submission Assignment 8; the supplied prompt labels its components Assignments #5 and #6.

T1. Specify the Blunder-Risk Model training-data source, blunder label, prediction-time features, and evaluation criteria with the advisor - **Owner: Braden Hayes.**

T2. Develop Authentication, Configuration, and Logging with Lichess sign-in, session checks, startup validation, and contextual error logs - **Owner: Nate Poppe.**

T3. Develop Game and Results Storage with database migrations, per-game transactions, duplicate prevention, and model-versioned scores - **Owner: Nate Poppe.**

T4. Obtain and prepare the Blunder-Risk Model training dataset with reproducible labels, missing-data rules, and leakage-controlled training, validation, and test splits - **Owner: Braden Hayes.**

T5. Develop Chess Feature Extraction with Stockfish analysis, reusable engine results, bounded parallel workers, and failure recovery - **Owner: Braden Hayes.**

T6. Develop Game Import and Validation for Lichess and PGN inputs with completed-game checks, progress tracking, duplicate prevention, and interrupted-import recovery - **Owner: Nate Poppe.**

T7. Develop and evaluate a baseline and candidate Blunder-Risk Model using the agreed validation criteria and probability-calibration checks - **Owner: Braden Hayes.**

T8. Develop the User Interface sign-in and import pages with input validation, availability counts, progress messages, and resume controls - **Owner: Nate Poppe.**

T9. Develop Blunder-Risk Model batch inference with feature-schema validation, missing-value preprocessing, versioned model loading, and isolated move failures - **Owner: Braden Hayes.**

T10. Develop the User Interface Games and Replay pages with the three highest-risk moves, exact board positions, risk scores, explanations, and engine-best moves - **Owner: Nate Poppe.**

T11. Develop Risk Explanation and Analytics with stored plain-language risk factors, rule-based fallbacks, and player-pattern aggregation after core explanations pass verification - **Owner: Braden Hayes.**

T12. Test User Interface, Game Import and Validation, and Game and Results Storage together for end-to-end workflows, interrupted imports, versioned results, and user-data isolation - **Owner: Nate Poppe.**

T13. Validate and refine Chess Feature Extraction, Blunder-Risk Model, and Risk Explanation and Analytics against held-out prediction results, reproducibility checks, explanation constraints, and the 20-game analysis target - **Owner: Braden Hayes.**

T14. Document and validate Configuration and User Interface deployment and operation with a reproducible local setup, demo dataset, and user guide - **Owner: Nate Poppe.**

Scope: analyze completed games only, following final D0-D2. The older project description uses the title above and refers to live scoring; reconcile that wording in C9/T1. All implementation tasks are planned, not already complete.

2. Milestones List - completed deliverables and current submission

All course deliverables preceding this planning submission (C1-C7) are complete. Canvas dates below are the course due dates, not actual completion dates.

ID / milestone	Date / basis / status	What it demonstrates	How it is verified	Verifier
C1 Professional Biography	Sep 02, 2026 Canvas Complete	Introduces each member and relevant skills.	Check both submitted biographies and Canvas receipts.	Instructor; each member
C2 Team Formation and Project Description	Sep 09, 2026 Canvas Complete	Records the two-person team and project purpose.	Compare the roster and supplied project-description.md with the submission.	Instructor; both members
C3 Advisor Selection and Design Constraints Essay	Sep 18, 2026 Canvas Complete	Identifies the advisor and the constraints governing design.	Check advisor confirmation and the submitted constraints essay.	Instructor; advisor
C4 User Stories and Use Cases	Sep 28, 2026 Canvas Complete	Defines the user workflow and acceptance criteria.	Check submitted stories and criteria referenced by D2.	Instructor; both members
C5 High-Level System Design (Design D0)	Oct 02, 2026 Canvas Complete	Defines six system modules, interfaces, and primary owners.	Inspect the supplied D0 artifact and its repository version.	Instructor; advisor
C6 Detailed System Design, Part 1 (Design D1)	Oct 08, 2026 Canvas Complete	Defines storage, import, feature extraction, and API contracts.	Inspect the supplied D1 artifact and its diagrams.	Instructor; advisor
C7 Detailed System Design, Part 2 (Design D2) and Peer Review Notes	Oct 14, 2026 Canvas Complete	Closes the design with model/UI behavior, edge cases, and review responses.	Inspect supplied D2 and its peer-review section; confirm separate notes in the repo.	Instructor; advisor
C8 Task List, Milestones, Timeline, and Effort Matrix	Oct 28, 2026 Canvas Planned	Establishes this consistent, resourced delivery baseline.	Submit the six-part PDF; verify Tasklist.md and Planning sources in the repo.	Instructor; both members

2. Milestones List - remaining fall deliverables

Team target means an internal readiness date, not a published course deadline. The roadmap names ethics, communication, and oral sessions but supplies no deadlines. The repeated Requirements/Technical Specification item is counted once. No completion is claimed for these future items.

ID / milestone	Date / basis / status	What it demonstrates	How it is verified	Verifier
C9 Requirements Document and Technical Specification Document	Oct 30, 2026 Canvas Planned	Freezes completed-game scope and measurable system contracts.	Compare requirements to D0-D2; resolve live-score wording and record open ML decisions.	Instructor; advisor
C10 Test and Validation Plan Draft	Nov 04, 2026 Canvas Planned	Maps requirements to fixtures, tests, metrics, and acceptance evidence.	Review test cases for timing, resume, isolation, model versions, and held-out evaluation.	Instructor; both members
C11 Ethics review package (roadmap Week 11)	Nov 06, 2026 Team target Planned	Documents fair play, data handling, and limits of predicted probabilities.	Advisor reviews completed-game-only policy, data provenance, and uncertainty wording.	Advisor; both members
C12 Communication package (roadmap Week 12)	Nov 13, 2026 Team target Planned	Makes the project understandable through a short narrative and draft slides.	Nate checks usability narrative; Braden checks modeling claims; advisor reviews.	Advisor; both members
C13 Oral presentation readiness 1 (roadmap Week 13)	Nov 18, 2026 Team target Planned	Demonstrates a coherent explanation of the design and delivery plan.	Run a timed rehearsal and record feedback plus assigned corrections.	Advisor; both members
C14 Oral presentation readiness 2 (roadmap Week 14)	Nov 25, 2026 Team target Planned	Shows that rehearsal feedback has been incorporated.	Repeat the timed presentation; close major feedback items.	Instructor; advisor
C15 Week 14 fall report	Nov 25, 2026 Team target Planned	Preserves the design baseline and reproduces the current planning tables.	Check all six components, identical task IDs, updated status, and source/PDF agreement.	Instructor; advisor
C16 Project Advisor Assessments	Dec 08, 2026 Canvas Planned	Provides the advisor with evidence of fall design progress and readiness.	Prepare evidence by Dec 4; advisor completes the course assessment by Dec 8.	Advisor; instructor
F1 Modeling protocol agreed	Nov 13, 2026 Internal Planned	Resolves data source, label, pre-move feature boundary, and evaluation protocol.	Advisor reviews T1 decision log, sample rows, and proposed model-selection criteria.	Advisor; Nate Poppe

2. Milestones List - Spring 2027 implementation

S1-S8 are internal outcomes, spaced 1-3 weeks apart. Advisor participation is proposed; both members preserve checkable evidence even if an advisor review must be asynchronous. Official CS 5002 deliverables will be added when published.

ID / milestone	Date / basis / status	What it demonstrates	How it is verified	Verifier
S1 Startup and engine feasibility demonstrated	Jan 22, 2027 Internal Planned	Authentication/configuration/logging run; fixed settings produce features on sample PGNs.	Run T2/T5 smoke tests and early 20-game timing probe; record hardware and engine settings.	Both members; advisor reviews evidence
S2 Stored-game and training-data baseline ready	Feb 05, 2027 Internal Planned	Schema and a bounded training dataset support repeatable development.	Apply migrations on a fresh database; audit label samples, missingness, and split leakage.	Both members; advisor reviews data protocol
S3 Resumable import and candidate model demonstrated	Feb 19, 2027 Internal Planned	Validated games survive interruption and a candidate improves on the agreed baseline.	Inject an import interruption; inspect duplicate counts and validation metrics/calibration.	Both members; advisor reviews model evidence
S4 Import-to-score pipeline demonstrated	Mar 05, 2027 Internal Planned	The import UI connects to versioned inference and displays correct progress.	Run completed-game import through stored scores; reject incompatible feature batches.	Both members
S5 Pre-break baseline frozen	Mar 12, 2027 Internal Planned	A tagged working baseline can be resumed after midterms and spring break.	Run smoke tests, back up DB/model artifacts, and document outstanding defects.	Both members
S6 Replay and explanations demonstrated	Apr 02, 2027 Internal Planned	Replay shows the selected board, top-risk moves, and readable factors.	Verify exact FENs and at least two supported factors of at most 25 words on acceptance fixtures.	Both members; advisor reviews demonstration
S7 Acceptance candidate approved	Apr 16, 2027 Internal Planned	Core requirements pass integrated, statistical, and performance checks.	Run the acceptance suite and held-out evaluation; time 20 games against 180 seconds; close release blockers.	Both members; advisor reviews evidence
S8 Final implementation and handoff ready	Apr 23, 2027 Internal Planned	A repeatable local release, user guide, evaluation report, and demo are ready before finals.	Install on a clean environment, rehearse the demo, verify artifact hashes and documentation.	Advisor; both members

3. Table 1 - task timeline (T1-T7)

Dates include each task's 20% effort contingency. E is the base team effort in hours; the allowance is 0.20E. Finish-to-start dependencies apply unless stated. Calendar windows are 1-2 weeks; tasks do not span winter or spring break.

Task	Exact task statement	Primary owner	Start date	Finish date	Depends on	Effort (h) base + 20%	Schedule path / risks
T1	Specify the Blunder-Risk Model training-data source, blunder label, prediction-time features, and evaluation criteria with the advisor.	Braden Hayes	Oct 29, 2026	Nov 10, 2026	C7	12 + 2.4 = 14.4	Prerequisite R1, R2
T2	Develop Authentication, Configuration, and Logging with Lichess sign-in, session checks, startup validation, and contextual error logs.	Nate Poppe	Jan 11, 2027	Jan 20, 2027	C7	14 + 2.8 = 16.8	App path R3, R7
T3	Develop Game and Results Storage with database migrations, per-game transactions, duplicate prevention, and model-versioned scores.	Nate Poppe	Jan 25, 2027	Feb 03, 2027	T2	14 + 2.8 = 16.8	App path R7
T4	Obtain and prepare the Blunder-Risk Model training dataset with reproducible labels, missing-data rules, and leakage-controlled training, validation, and test splits.	Braden Hayes	Jan 25, 2027	Feb 03, 2027	T1, T5	16 + 3.2 = 19.2	Model path R1, R2
T5	Develop Chess Feature Extraction with Stockfish analysis, reusable engine results, bounded parallel workers, and failure recovery.	Braden Hayes	Jan 11, 2027	Jan 20, 2027	T1	16 + 3.2 = 19.2	Model path R4
T6	Develop Game Import and Validation for Lichess and PGN inputs with completed-game checks, progress tracking, duplicate prevention, and interrupted-import recovery.	Nate Poppe	Feb 08, 2027	Feb 17, 2027	T2, T3	18 + 3.6 = 21.6	App path R3, R7
T7	Develop and evaluate a baseline and candidate Blunder-Risk Model using the agreed validation criteria and probability-calibration checks.	Braden Hayes	Feb 08, 2027	Feb 17, 2027	T4	16 + 3.2 = 19.2	Model path R1, R2

T5 initially runs against local fixtures; T4 uses offline files, so neither waits for application storage. T1 records the completed-game scope decision by Oct 29 for C9; its complete modeling protocol is due Nov 10.

3. Table 1 - task timeline (T8-T14)

Dates include each task's 20% effort contingency. E is the base team effort in hours; the allowance is 0.20E. Finish-to-start dependencies apply unless stated. Calendar windows are 1-2 weeks; tasks do not span winter or spring break.

Task	Exact task statement	Primary owner	Start date	Finish date	Depends on	Effort (h) base + 20%	Schedule path / risks
T8	Develop the User Interface sign-in and import pages with input validation, availability counts, progress messages, and resume controls.	Nate Poppe	Feb 22, 2027	Mar 03, 2027	T2, T6	14 + 2.8 = 16.8	App path R3
T9	Develop Blunder-Risk Model batch inference with feature-schema validation, missing-value preprocessing, versioned model loading, and isolated move failures.	Braden Hayes	Feb 22, 2027	Mar 03, 2027	T3, T5, T7	15 + 3.0 = 18.0	Model path R2, R7
T10	Develop the User Interface Games and Replay pages with the three highest-risk moves, exact board positions, risk scores, explanations, and engine-best moves.	Nate Poppe	Mar 22, 2027	Mar 31, 2027	T3, T8, T9	16 + 3.2 = 19.2	App path R5, R7
T11	Develop Risk Explanation and Analytics with stored plain-language risk factors, rule-based fallbacks, and player-pattern aggregation after core explanations pass verification.	Braden Hayes	Mar 22, 2027	Mar 31, 2027	T9	15 + 3.0 = 18.0	Model path R2, R5
T12	Test User Interface, Game Import and Validation, and Game and Results Storage together for end-to-end workflows, interrupted imports, versioned results, and user-data isolation.	Nate Poppe	Apr 05, 2027	Apr 14, 2027	T6, T8, T10, T11	16 + 3.2 = 19.2	App path R3, R7
T13	Validate and refine Chess Feature Extraction, Blunder-Risk Model, and Risk Explanation and Analytics against held-out prediction results, reproducibility checks, explanation constraints, and the 20-game analysis target.	Braden Hayes	Apr 05, 2027	Apr 14, 2027	T5, T7, T9, T10, T11	16 + 3.2 = 19.2	Model path R2, R4
T14	Document and validate Configuration and User Interface deployment and operation with a reproducible local setup, demo dataset, and user guide.	Nate Poppe	Apr 19, 2027	Apr 21, 2027	T12, T13, S7	10 + 2.0 = 12.0	Release path R6, R7, R8

T10 and T11 run in parallel against the D2 response schema and shared fixtures. T12 waits for both real implementations. Optional player patterns in T11 are the first scope reduction if core acceptance slips.

3. Table 1 - dated milestone rows

Each milestone in Section 2 appears here exactly once. Historical due dates are preserved. Buffer IDs refer to the next table; historical work is not assigned retrospective buffer.

ID	Milestone outcome	Completion / due date	Date basis	Status	Protected block
C1	Professional Biography	Sep 02, 2026	Canvas	Complete	Historical
C2	Team Formation and Project Description	Sep 09, 2026	Canvas	Complete	Historical
C3	Advisor Selection and Design Constraints Essay	Sep 18, 2026	Canvas	Complete	Historical
C4	User Stories and Use Cases	Sep 28, 2026	Canvas	Complete	Historical
C5	High-Level System Design (Design D0)	Oct 02, 2026	Canvas	Complete	Historical
C6	Detailed System Design, Part 1 (Design D1)	Oct 08, 2026	Canvas	Complete	Historical
C7	Detailed System Design, Part 2 (Design D2) and Peer Review Notes	Oct 14, 2026	Canvas	Complete	Historical
C8	Task List, Milestones, Timeline, and Effort Matrix	Oct 28, 2026	Canvas	Planned	B0
C9	Requirements Document and Technical Specification Document	Oct 30, 2026	Canvas	Planned	B1
C10	Test and Validation Plan Draft	Nov 04, 2026	Canvas	Planned	B2
C11	Ethics review package (roadmap Week 11)	Nov 06, 2026	Team target	Planned	B3
C12	Communication package (roadmap Week 12)	Nov 13, 2026	Team target	Planned	B4
C13	Oral presentation readiness 1 (roadmap Week 13)	Nov 18, 2026	Team target	Planned	B5
C14	Oral presentation readiness 2 (roadmap Week 14)	Nov 25, 2026	Team target	Planned	B6
C15	Week 14 fall report	Nov 25, 2026	Team target	Planned	B6
C16	Project Advisor Assessments	Dec 08, 2026	Canvas	Planned	B7
F1	Modeling protocol agreed	Nov 13, 2026	Internal	Planned	B4
S1	Startup and engine feasibility demonstrated	Jan 22, 2027	Internal	Planned	B8
S2	Stored-game and training-data baseline ready	Feb 05, 2027	Internal	Planned	B9
S3	Resumable import and candidate model demonstrated	Feb 19, 2027	Internal	Planned	B10
S4	Import-to-score pipeline demonstrated	Mar 05, 2027	Internal	Planned	B11
S5	Pre-break baseline frozen	Mar 12, 2027	Internal	Planned	B12
S6	Replay and explanations demonstrated	Apr 02, 2027	Internal	Planned	B13
S7	Acceptance candidate approved	Apr 16, 2027	Internal	Planned	B14
S8	Final implementation and handoff ready	Apr 23, 2027	Internal	Planned	B15

3. Table 1 - protected milestone blocks

These blocks are additional to task-level 20% contingency. Use them for corrective work, verification, and rehearsal; do not commit new features into them. Spring blocks reserve 1 hour per member within the weekly capacity budget, plus uncommitted time where available.

ID	Start	End	Supports	Purpose / protected activity
B0	Oct 28, 2026	Oct 28, 2026	C8	Final consistency and export check; no retrospective buffer is claimed.
B1	Oct 29, 2026	Oct 29, 2026	C9	Scope wording and requirement/specification review; T1 offers its scope decision before its full completion.
B2	Nov 02, 2026	Nov 03, 2026	C10	Review the test-plan draft; no new feature commitments in this block.
B3	Nov 05, 2026	Nov 05, 2026	C11	Review the ethics package and fair-play boundary.
B4	Nov 12, 2026	Nov 12, 2026	C12, F1	Review communication materials and the T1 modeling protocol after the Nov 11 holiday.
B5	Nov 16, 2026	Nov 17, 2026	C13	Presentation corrections and rehearsal only.
B6	Nov 23, 2026	Nov 24, 2026	C14, C15	Freeze tables, resolve report inconsistencies, rehearse; preserve Thanksgiving Nov 26-29.
B7	Dec 03, 2026	Dec 04, 2026	C16	Send-ready assessment evidence prepared before finals; no scheduled exam-week development.
B8	Jan 21, 2027	Jan 21, 2027	S1	Startup/engine fixes and evidence review only; reserve 1 hour per member.
B9	Feb 04, 2027	Feb 04, 2027	S2	Schema/data fixes and evidence review only; reserve 1 hour per member.
B10	Feb 18, 2027	Feb 18, 2027	S3	Import/model corrections only; reserve 1 hour per member.
B11	Mar 04, 2027	Mar 04, 2027	S4	Integration fixes only; reserve 1 hour per member.
B12	Mar 11, 2027	Mar 11, 2027	S5	Backup/recovery checks and defects only; reserve 1 hour per member.
B13	Apr 01, 2027	Apr 01, 2027	S6	Replay/explanation fixes only; reserve 1 hour per member.
B14	Apr 15, 2027	Apr 15, 2027	S7	Release-blocking fixes only; reserve 1 hour per member.
B15	Apr 22, 2027	Apr 22, 2027	S8	Clean-install/demo corrections only; reserve 1 hour per member.

3. Table 1 - availability, breaks, and critical paths

ID	Start	End	Constraint	Schedule response
K1	Nov 26, 2026	Nov 29, 2026	Thanksgiving blackout	No planned project work.
K2	Dec 06, 2026	Dec 12, 2026	Fall exams	No development; advisor assessment remains due Dec 8 using evidence prepared Dec 4.
K3	Dec 13, 2026	Jan 10, 2027	Winter blackout	No planned development; restart Jan 11.
K4	Jan 18, 2027	Jan 18, 2027	University holiday	No required work; T2/T5 budgets fit the remaining days.
K5	Mar 08, 2027	Mar 12, 2027	Assumed midterm reserve	Capacity reduced to 6 hours per member; no new tasks; freeze and repair existing work.
K6	Mar 15, 2027	Mar 21, 2027	Spring break blackout	No planned project work; T10/T11 start after the break.
K7	Apr 24, 2027	Apr 29, 2027	Spring finals blackout	No planned development; release ready Apr 23.
K8	Jan 11, 2027	Apr 23, 2027	Spring availability constraint	Braden: 18 credits. Nate: expected 15 credits plus an assumed 20 co-op hours/week. Budget 12 project hours/week each, including coordination and reserves.

Controlling delivery paths

Model path: T5 -> T4 -> T7 -> T9 -> T11 -> T13 -> T14 -> S8. App path: T2 -> T3 -> T6 -> T8 -> T10 -> T12 -> T14 -> S8. T1 is the fall prerequisite for the model path; T9 also feeds T10, and both T10/T11 feed validation.

These are the critical, resource-constrained paths for the fixed milestone dates. Core tasks on either path cannot slip past their gate without consuming a protected block, reducing optional patterns, or replanning a downstream date. Midterm and holiday blackouts are not usable schedule float.

Proposed capacity: 12 project hours/week per member, including 1 hour of coordination. Braden has 18 spring credits; Nate is expected to take 15 credits alongside part-time co-op, estimated at 20 hours/week. Midterm capacity is 6 hours/week each; break capacity is zero. Review actual availability weekly; these are planning assumptions, not reported commitments.

4. Table 2 - effort matrix (hours)

Hours include each member's proportional 20% contingency, already shown in Table 1; do not add it a second time. P marks the single primary owner, who always carries the largest share. Estimates are proposed planning budgets, not measured work or owner-elicited PERT estimates.

Task	Exact task statement	Braden Hayes	Nate Poppe	Row total
T1	Specify the Blunder-Risk Model training-data source, blunder label, prediction-time features, and evaluation criteria with the advisor.	12.0 P	2.4	14.4
T2	Develop Authentication, Configuration, and Logging with Lichess sign-in, session checks, startup validation, and contextual error logs.	2.4	14.4 P	16.8
T3	Develop Game and Results Storage with database migrations, per-game transactions, duplicate prevention, and model-versioned scores.	2.4	14.4 P	16.8
T4	Obtain and prepare the Blunder-Risk Model training dataset with reproducible labels, missing-data rules, and leakage-controlled training, validation, and test splits.	16.8 P	2.4	19.2
T5	Develop Chess Feature Extraction with Stockfish analysis, reusable engine results, bounded parallel workers, and failure recovery.	16.8 P	2.4	19.2
T6	Develop Game Import and Validation for Lichess and PGN inputs with completed-game checks, progress tracking, duplicate prevention, and interrupted-import recovery.	3.6	18.0 P	21.6
T7	Develop and evaluate a baseline and candidate Blunder-Risk Model using the agreed validation criteria and probability-calibration checks.	16.8 P	2.4	19.2
T8	Develop the User Interface sign-in and import pages with input validation, availability counts, progress messages, and resume controls.	2.4	14.4 P	16.8
T9	Develop Blunder-Risk Model batch inference with feature-schema validation, missing-value preprocessing, versioned model loading, and isolated move failures.	14.4 P	3.6	18.0
T10	Develop the User Interface Games and Replay pages with the three highest-risk moves, exact board positions, risk scores, explanations, and engine-best moves.	2.4	16.8 P	19.2
T11	Develop Risk Explanation and Analytics with stored plain-language risk factors, rule-based fallbacks, and player-pattern aggregation after core explanations pass verification.	14.4 P	3.6	18.0
T12	Test User Interface, Game Import and Validation, and Game and Results Storage together for end-to-end workflows, interrupted imports, versioned results, and user-data isolation.	4.8	14.4 P	19.2
T13	Validate and refine Chess Feature Extraction, Blunder-Risk Model, and Risk Explanation and Analytics against held-out prediction results, reproducibility checks, explanation constraints, and the 20-game analysis target.	15.6 P	3.6	19.2
T14	Document and validate Configuration and User Interface deployment and operation with a reproducible local setup, demo dataset, and user guide.	2.4	9.6 P	12.0
TOTAL	All 14 tasks	127.2	122.4	249.6

Base effort: 208 team hours; task contingency: 41.6 hours; total: 249.6 hours. Braden carries 4.8 more hours (3.8% of the average load). If his course load constrains delivery, move up to 2.4 hours of T13 test execution from Braden to Nate; Braden retains the largest share and primary ownership. If Nate instead loses availability, move T12 fixture work to Braden within the same ownership rule.

4. Effort feasibility and estimation notes

Work window (2027)	Tasks	Braden hours	Nate hours	Capacity each	Coordination + gate reserve
Jan 11-22	T2, T5	22.2	19.8	24	3 h/member included
Jan 25-Feb 5	T3, T4	22.2	19.8	24	3 h/member included
Feb 8-19	T6, T7	23.4	23.4	24	3 h/member included
Feb 22-Mar 5	T8, T9	19.8	21.0	24	3 h/member included
Mar 22-Apr 2	T10, T11	19.8	23.4	24	3 h/member included
Apr 5-16	T12, T13	23.4	21.0	24	3 h/member included
Apr 19-23	T14	4.4	11.6	12	2 h/member included

How to interpret the hours

The windows include buffered task work, one coordination hour per week, and one protected gate hour per member. Allocate work unevenly within each window if needed to stay below the weekly cap; task work ends before its protected gate. This is a bounded first implementation, using the designed libraries and a limited training sample, not open-ended model research.

The matrix covers the numbered implementation tasks. Separate planning allowances cover fall coursework (24 hours per member across the remaining fall period), spring coordination (14 hours per member), and eight spring gate reserves (8 hours per member). The midterm freeze/backup review uses at most 2 additional hours per member. These allowances are not hidden in task totals or added as duplicate T-rows.

Scope control: settle the label and dataset in T1; build one baseline and one candidate in T7; complete risk factors before optional player patterns in T11. If the 20% contingency is exhausted, record the forecast, drop optional scope first, and obtain an agreed rebaseline rather than claiming the old deadline is still achievable.

Review rule: update actual effort and remaining estimates weekly. If any two-week package exceeds 24 hours for either person, split or re-scope it and update all four task-bearing artifacts together. Retain one primary owner per task.

5. Work breakdown structure

Level 1 = phase; level 2 = numbered tasks. IDs, statements, and owners match the Task List, timeline, and effort matrix. Phase membership is organizational; dependencies determine execution order.

1. Scope and foundations

T1. Specify the Blunder-Risk Model training-data source, blunder label, prediction-time features, and evaluation criteria with the advisor - Owner: Braden Hayes.

T2. Develop Authentication, Configuration, and Logging with Lichess sign-in, session checks, startup validation, and contextual error logs - Owner: Nate Poppe.

T3. Develop Game and Results Storage with database migrations, per-game transactions, duplicate prevention, and model-versioned scores - Owner: Nate Poppe.

2. Data and model development

T4. Obtain and prepare the Blunder-Risk Model training dataset with reproducible labels, missing-data rules, and leakage-controlled training, validation, and test splits - Owner: Braden Hayes.

T5. Develop Chess Feature Extraction with Stockfish analysis, reusable engine results, bounded parallel workers, and failure recovery - Owner: Braden Hayes.

T6. Develop Game Import and Validation for Lichess and PGN inputs with completed-game checks, progress tracking, duplicate prevention, and interrupted-import recovery - Owner: Nate Poppe.

T7. Develop and evaluate a baseline and candidate Blunder-Risk Model using the agreed validation criteria and probability-calibration checks - Owner: Braden Hayes.

3. Scoring and user experience

T8. Develop the User Interface sign-in and import pages with input validation, availability counts, progress messages, and resume controls - Owner: Nate Poppe.

T9. Develop Blunder-Risk Model batch inference with feature-schema validation, missing-value preprocessing, versioned model loading, and isolated move failures - Owner: Braden Hayes.

T10. Develop the User Interface Games and Replay pages with the three highest-risk moves, exact board positions, risk scores, explanations, and engine-best moves - Owner: Nate Poppe.

T11. Develop Risk Explanation and Analytics with stored plain-language risk factors, rule-based fallbacks, and player-pattern aggregation after core explanations pass verification - Owner: Braden Hayes.

4. Validation and delivery

T12. Test User Interface, Game Import and Validation, and Game and Results Storage together for end-to-end workflows, interrupted imports, versioned results, and user-data isolation - Owner: Nate Poppe.

T13. Validate and refine Chess Feature Extraction, Blunder-Risk Model, and Risk Explanation and Analytics against held-out prediction results, reproducibility checks, explanation constraints, and the 20-game analysis target - Owner: Braden Hayes.

T14. Document and validate Configuration and User Interface deployment and operation with a reproducible local setup, demo dataset, and user guide - Owner: Nate Poppe.

6. Risk register - technical and external risks

Likelihood: low <25%, medium 25-60%, high >60% (planning judgments). High impact threatens a core requirement or milestone; medium impact consumes contingency or optional scope. Owners review triggers weekly and record actions.

R1 | Training data, labels, or advisor decision arrive late

Category: Technical / external | Likelihood: Medium | Impact: High | Strategy: Reduce | Owner: Braden Hayes

Mitigation: Limit the first dataset to a reproducible sample; agree the blunder label and evaluation protocol in T1; retain a simple baseline rather than extending model research.

Trigger: No usable labeled sample or advisor decision by F1 (Nov 13).

Visible in Table 1: T1, T4, T7; F1, S2; B4, B9

R2 | Leakage, missing metadata, or weak calibration invalidates predictions

Category: Technical | Likelihood: Medium | Impact: High | Strategy: Reduce | Owner: Braden Hayes

Mitigation: Keep played-move outcome information out of input features; split by game and time, add player-disjoint evaluation if claiming new-player generalization, and audit imputation and calibration.

Trigger: Leakage found, or candidate fails agreed validation criteria at S3.

Visible in Table 1: T4, T7, T9, T13; B10, B14

R3 | Lichess sign-in, rate limits, or resume semantics block import

Category: External | Likelihood: Medium | Impact: High | Strategy: Reduce | Owner: Nate Poppe

Mitigation: Verify sign-in and API behavior early; preserve each saved game; test interruption with fixtures; retain PGN input for a demo while fixing API failures.

Trigger: Sign-in cannot complete at S1, or resume duplicates/re-downloads games at S3.

Visible in Table 1: T2, T6, T8, T12; B8, B10

R4 | Stockfish exceeds the 180-second target

Category: Technical | Likelihood: High | Impact: High | Strategy: Reduce | Owner: Braden Hayes

Mitigation: Measure early in T5; pin engine/settings, reuse results, cap workers, and tune search while checking feature quality; recheck the full workflow in T13.

Trigger: Early 20-game probe exceeds 180 seconds or projects an overrun at S1.

Visible in Table 1: T5, T13; S1, S7; B8, B14

6. Risk register - team, integration, and schedule risks

Likelihood: low <25%, medium 25-60%, high >60% (planning judgments). High impact threatens a core requirement or milestone; medium impact consumes contingency or optional scope. Owners review triggers weekly and record actions.

R5 | Optional patterns or model experiments displace core functionality

Category: Team / scope | Likelihood: Medium | Impact: Medium | Strategy: Avoid | Owner: Braden Hayes

Mitigation: Deliver factors and core replay first; limit model comparison to one baseline and one candidate; defer player patterns if the core path slips.

Trigger: Any core task uses all its contingency, or factors/replay are incomplete by Mar 29.

Visible in Table 1: T7, T10, T11; S6; B13

R6 | Course load, co-op shifts, exams, or breaks reduce availability

Category: Team | Likelihood: High | Impact: High | Strategy: Reduce | Owner: Nate Poppe

Mitigation: Account for Braden's 18 credits and Nate's expected 15 credits plus part-time co-op; use 12-hour project caps, one-hour coordination, cross-review, and protected breaks; review availability weekly and transfer support work before touching milestone buffers.

Trigger: Either member forecasts under 8 available hours for two successive working weeks.

Visible in Table 1: K1-K8; B8-B15; all task contingency

R7 | Integration defects, lost results, or environment differences delay release

Category: Technical / team | Likelihood: Medium | Impact: High | Strategy: Reduce | Owner: Nate Poppe

Mitigation: Share API/feature fixtures before parallel UI/explanation work; test migrations, versioned scores, user isolation, backups, and a clean local deployment.

Trigger: Contract tests fail at S4/S6, restored data differ, or clean install fails before S8.

Visible in Table 1: T3, T9-T14; B11, B13-B15

R8 | Unpublished course deadlines or assessment requirements change

Category: External | Likelihood: Medium | Impact: High | Strategy: Reduce | Owner: Nate Poppe

Mitigation: Check Canvas weekly; replace provisional dates as soon as posted; keep internal release Apr 23 and move work earlier if the official demo is earlier.

Trigger: New official deadline is earlier than a team target, or a new graded item is posted.

Visible in Table 1: C11-C15 and S8 provisional dates; B3-B6, B15

Source and status notes

Project title: supplied project-description.md. Module names, ownership, behavior, and acceptance targets: supplied D0, D1, and D2 PDFs. Course due dates: supplied Canvas module listing; topic sequence: supplied 14-week roadmap. C1-C7 are complete; C11-C15 dates remain internal targets. Repository commits are managed separately from this package.

Academic dates: University of Cincinnati, 2026-2027 Academic Year Calendar (checked Oct 28, 2026). Spring classes: Jan 11-Apr 23; spring break: Mar 15-21; spring finals: Apr 24-29. The full winter project blackout and midterm reserve are team planning choices.